



Wood Surface Defect Detection

Using YOLOv8n with Class-Aware Augmentation

Computer Vision Project — University of Engineering & Technology, Lahore



Problem Definition

Wood surface defects — **quartzity, knots, cracks, resin pockets, and marrow** — significantly degrade timber product quality. Manual inspection is subjective, slow, and error-prone. This project delivers an automated, real-time detection system using a fine-tuned **YOLOv8n** deep learning model, enabling consistent quality control on production lines.

Augmentation Strategy

Class-Aware Oversampling

Minority classes augmented to match median (412/class):

✓ H-Flip / V-Flip

✓ $\pm 10^\circ$ Rotation

✓ $\pm 15\%$ Brightness/Contrast

✓ Min visibility 30%

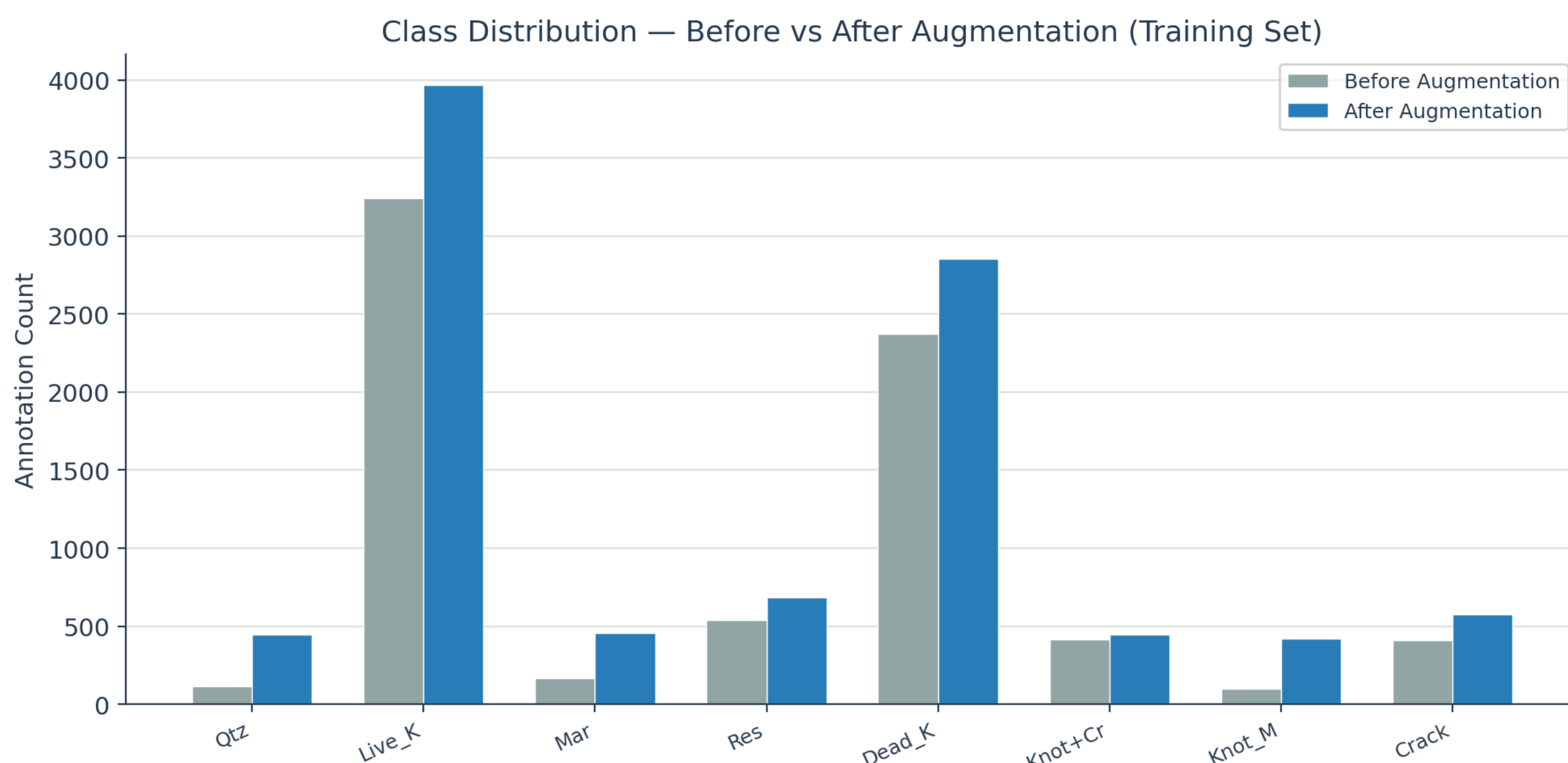
Technologies

FastAPI + Bootstrap 5

Full-stack web application for real-time edge deployment:

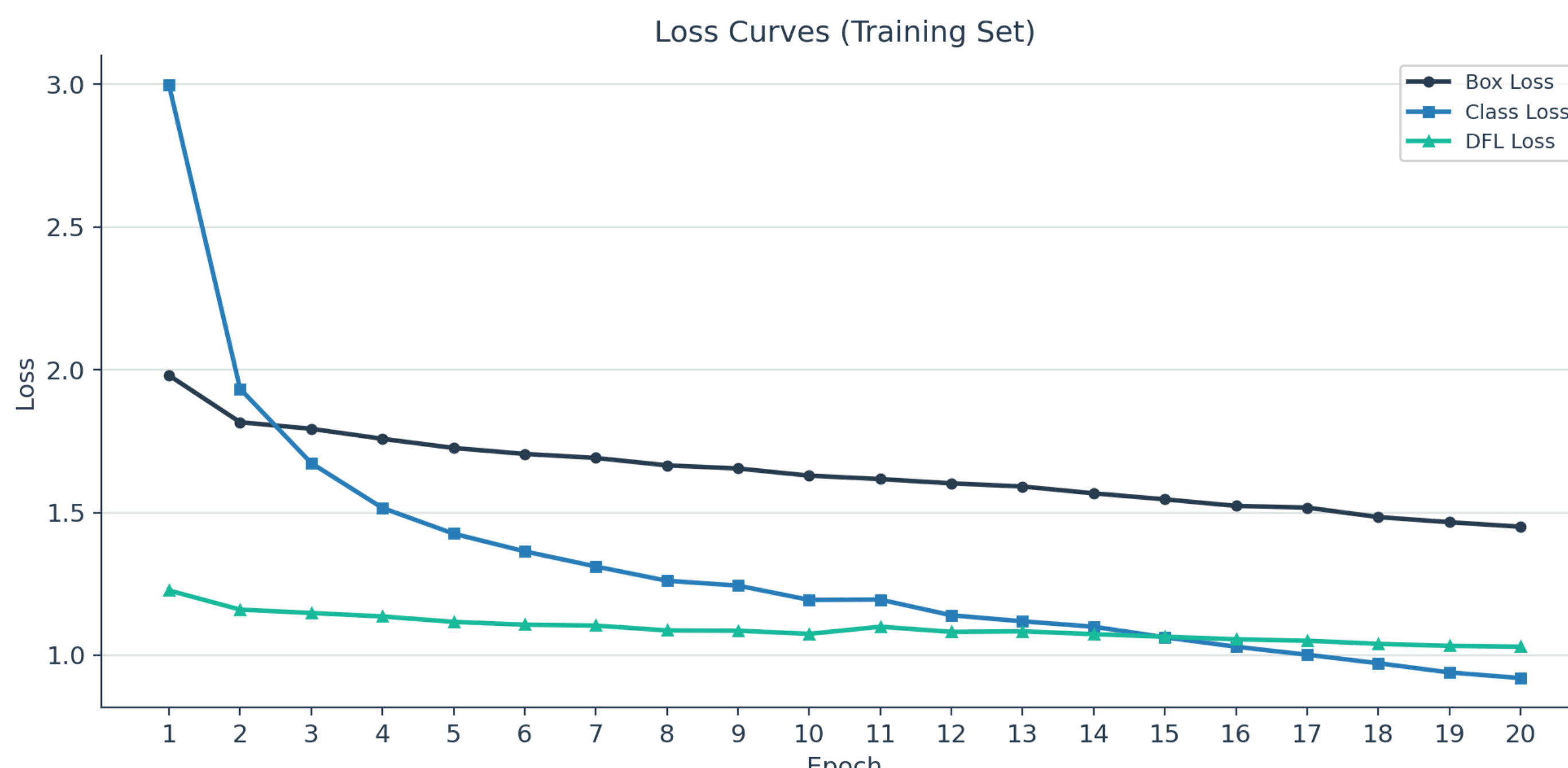
- Upload images for single-shot detection
- Live webcam inference pipeline
- Color-coded class legend with confidence scores
- NMS + confidence threshold filtering

Class Imbalance (Before Augmentation)



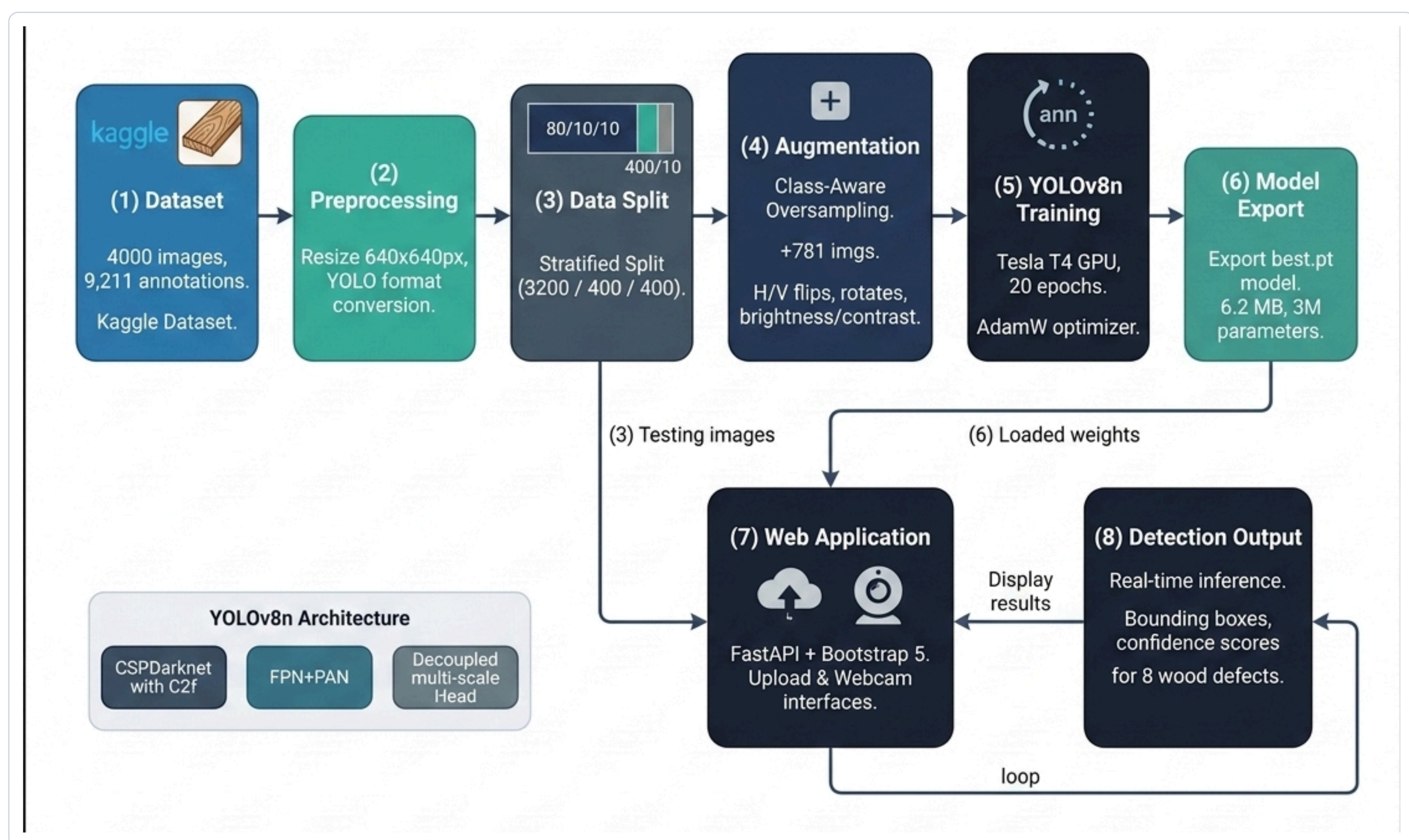
Quartzity (116), Marrow (166), Knot_missing (97) severely underrepresented → balanced to 420-445 via augmentation.

Training Progress — Loss Curves



Box, class, and DFL losses converge steadily over 20 epochs. Tesla T4 GPU (0.684 hrs). No overfitting observed.

Architecture



Conclusion

- YOLOv8n achieves **mAP@50 of 0.639** across 8 defect classes
- Class-aware augmentation boosted minority recall **3-4x**
- FastAPI web app optimized for low-latency deployment
- **Best:** Dead_Knot (0.852)
- **Worst:** Quartzity (0.155)

Key Metrics

- **mAP@50:** 0.639
- **Precision:** 0.665
- **Recall:** 0.643
- **Training:** 0.684 hrs (Tesla T4)
- **Epochs:** 20

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